Tracking Public Interest in Temperature Change: A Time Series Analysis of Google Trends Data

By: Joseph Cha and Wilson Cam

Abstract

Public awareness and concern about climate change have grown significantly in recent years, often reflected in online search behavior. This study conducts a time series analysis of Google Trends data on searches related to “temperature change” to examine patterns, seasonality, and long-term trends in public interest. The dataset is first preprocessed by identifying and removing outliers before being transformed and analyzed for stationarity using the Augmented Dickey-Fuller Test. Time series decomposition is performed to isolate trend and seasonal components, while ARIMA and SARIMA models are fitted to forecast future search interest. Results indicate a strong seasonal pattern, suggesting that public interest in temperature change is influenced by recurring climate events. This study analyzes Google Trends data on public search interest in ‘temperature change’ using advanced time series methods. Through stationarity tests, time series decomposition, and ARIMA/SARIMA modeling, we uncover long-term trends, seasonal effects, and the predictive potential of search interest in climate-related topics.

Introduction

The term *"temperature change"* is widely used in discussions about climate variability, global warming, and weather patterns. In recent years, climate change and environmental shifts have become pressing global concerns, leading to fluctuations in public interest in temperature change. Understanding the search term variations over time can give us valuable insights into public awareness, engagement, and responsiveness to climate-related issues.

This study aims to analyze Google Trends data on search interest in “temperature change” using time series statistical methods. By applying techniques such as stationarity tests, time series decomposition, and ARIMA/SARIMA modeling, we examine underlying trends, seasonal effects, and the potential for forecasting future public interest in climate-related searches. This research builds upon existing studies that use online search behavior to track public awareness of environmental and social issues, contributing to the understanding of how digital data can serve as an indicator of societal concern.

The remainder of this paper is structured to provide a comprehensive analysis of public interest in temperature change using time series methods. We first introduce the dataset, describing its source, structure, and key characteristics. It also outlines preprocessing steps, including outlier detection and necessary transformations to prepare the data for analysis. We next present the core data analysis, applying statistical tests to assess stationarity, performing time series decomposition to examine trends and seasonality, and developing ARIMA and SARIMA models for forecasting future search interest. Then we discuss the key findings, interpreting the observed trends and patterns while addressing the implications of fluctuating public interest in climate-related topics. Additionally, this section critically evaluates the limitations of the study

and challenges faced during the modeling process. We finally conclude the paper by summarizing the main insights derived from the analysis and suggesting directions for future research. At the end, we include the complete code and supplementary calculations used in the project.

Data Description

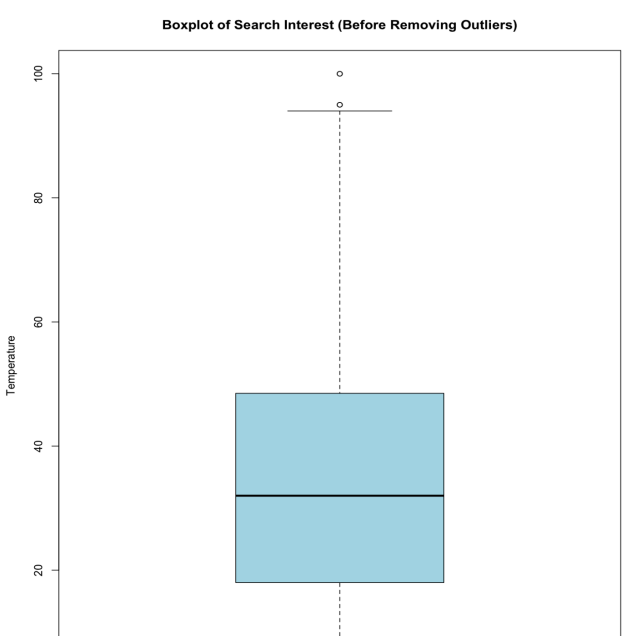
The dataset used in this study is obtained from Google Trends, which provides normalized search interest data specific to the term “Temperature Change”. Google Trends provides a real-time measure of public search behavior, making it a valuable tool for tracking shifts in awareness on climate-related topics. The information is based on a 0 to 100 relative scale, where 100 indicates the highest search interest for the term in the selected timeframe (in this case 2004 to present day) and lower values represented relative to that peak. By examining the dataset, we can identify trends, seasonal components, and fluctuations in the public interest related to “Temperature Change”.

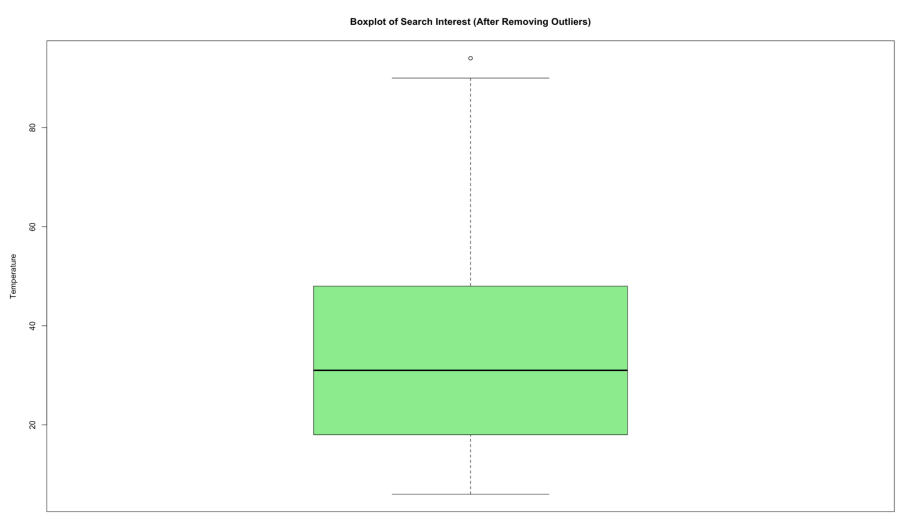
Summary Statistics:

Minimum	1st Quartile	Median	Mean	3rd Quartile	Maximum
6.00	18.00	31.00	35.83	48.00	94.00

Taking a look at our summary statistics, we are able to gain several insights. We can see the range goes from 6 to 94, indicating a large spread across the data. We also observe a slightly right skewed distribution since the mean of 35.83 is larger than the mean of 31. Additionally, we see that 50% of the data is spread between search interest values of 18 and 48 (30 units). Finally, we can identify a potential outlier in line with the likely right skewed distribution. 94 is significantly larger than 48 so we should be prepared to make some adjustments to our data before it is ready for further analysis.

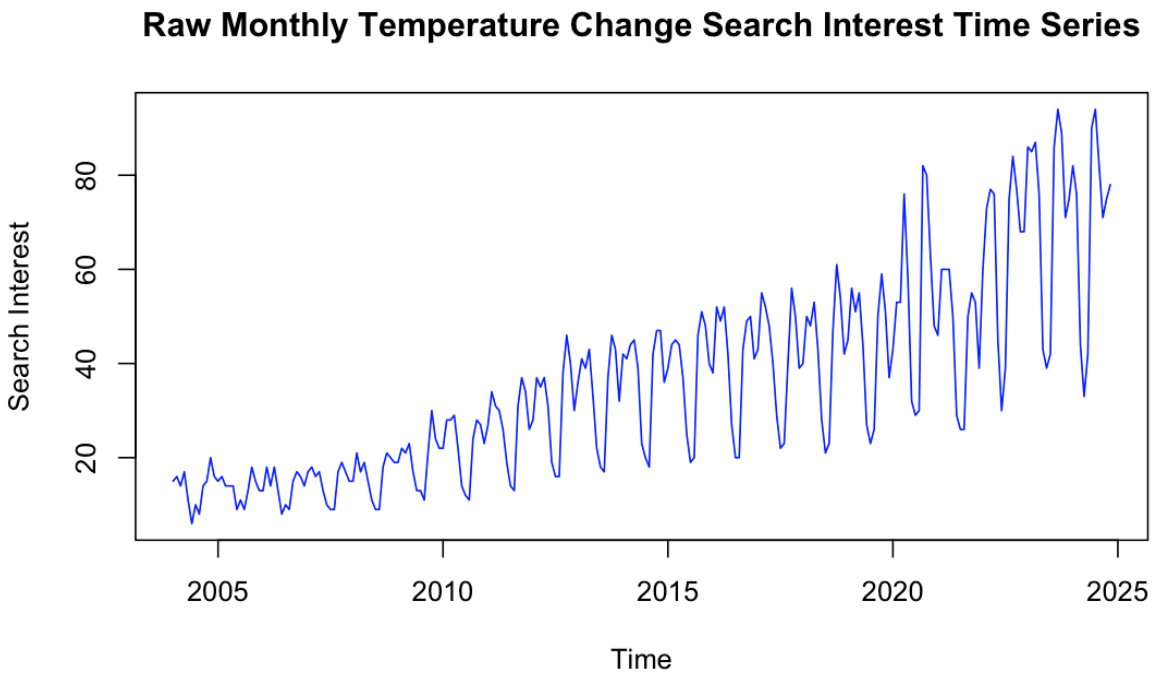
Boxplots for Outlier Detection:





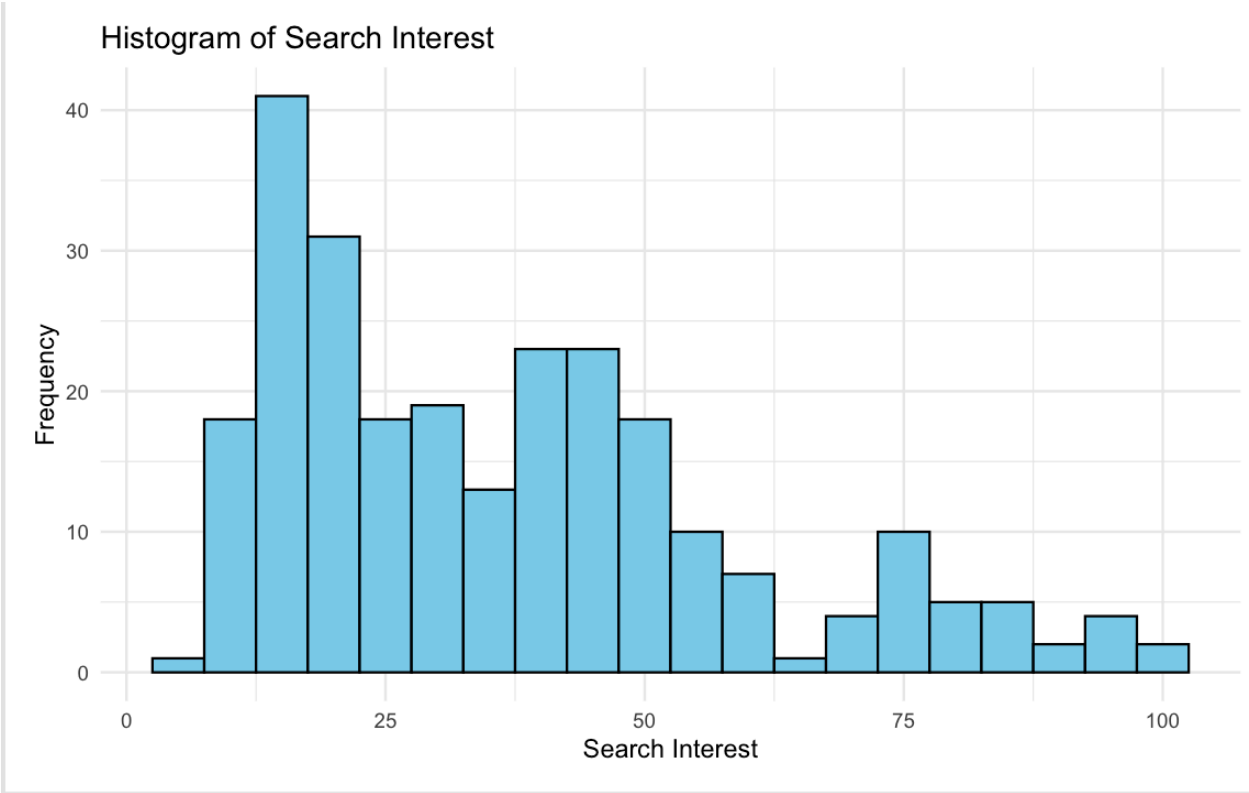
The plots represent the boxplot distribution of the data before (left graph) and after (right graph) outlier removal. We had a total of 4 outliers and after removal, we can see that the data now has a more balanced distribution. The whiskers are shorter and the extreme high values that were driving much of the variation in the original dataset are now gone. Outliers were removed rather than adjusted because extreme peaks in search data may result from short-lived viral events rather than meaningful long-term trends.

Raw Time Series Plot:

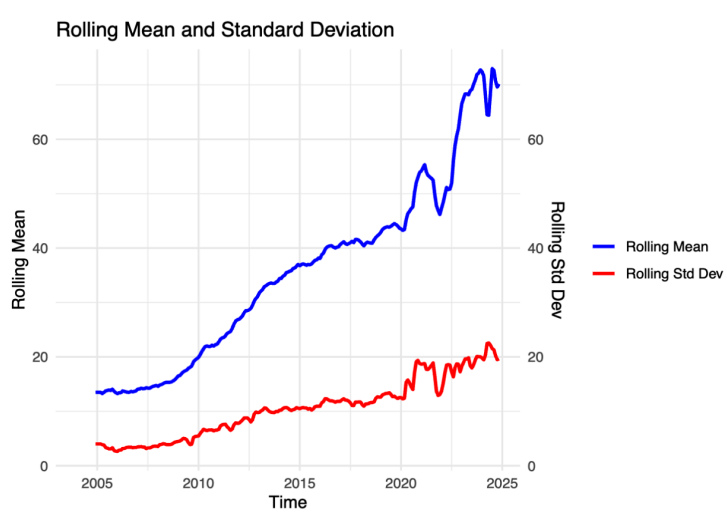


Through the raw plot of our data, we are able to draw several conclusions. There is a definite upward trend present in the data, meaning that for whatever reason (media coverage, articles, or broader societal concern) , the interest in temperature change is increasing online. We also clearly see seasonality present in the data, perhaps an annual seasonal fluctuation due to specific events such as summer or winter related searches. The fluctuations are also increasing in their magnitudes as the years progress, indicating heteroscedasticity (changing variance). This suggests that variations in search interest are becoming more pronounced as time progresses, meaning that extreme highs and lows are more common in later years. To address this, a log transformation may be needed.

Histogram:



This histogram is just another perspective of the raw data before any data transformation. With the previous observations in mind, we are additionally able to see a potential bimodal distribution present with the peaks forming around 15-20, suggesting that a significant number of instances fall within this lower range, while another concentration occurs around 40-50, representing a second common search interest level.

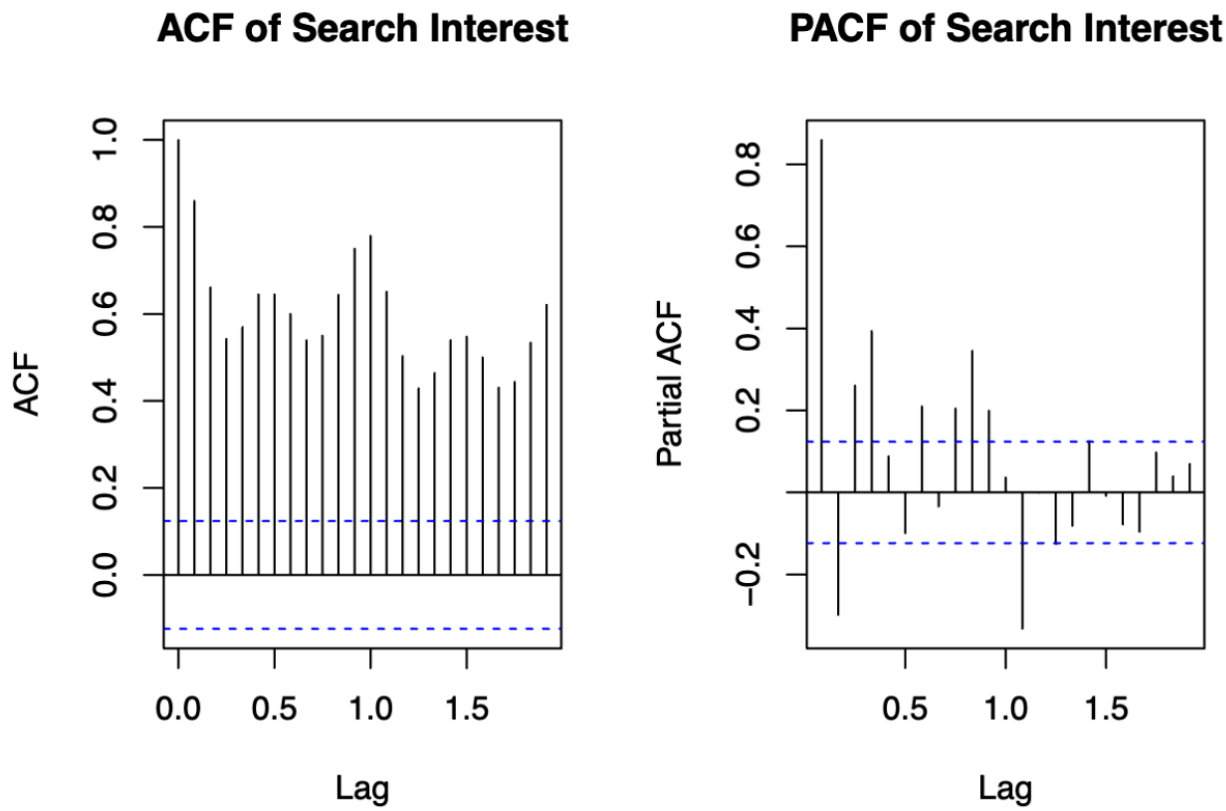


Rolling Mean and Rolling Standard Deviation:

From the mean and rolling standard deviation, we see in the earlier periods of years 2005 to 2015, both the rolling mean and standard deviation remain relatively stable, but fluctuations start increasing

around 2015. Then from 2019 onward, there is a sharp increase in both the rolling mean and standard deviation, suggesting significant changes in the data. Additionally, we see the presence of potential heteroscedasticity once more with the increasing variability over time. Without constant variability, future predictions might be less reliable if this is not addressed. Once again, a log transformation may need to be applied.

ACF and PACF:



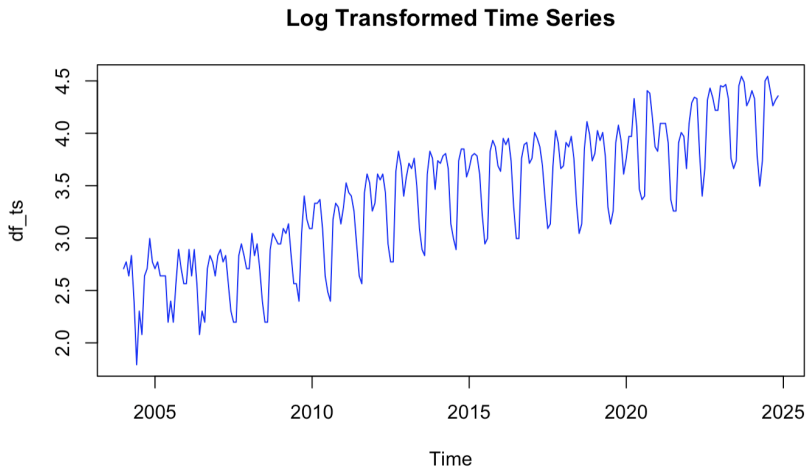
The ACF plot shows strong autocorrelation with a slow decay, indicating that the time series is non-stationary and requires differencing (d=1) to remove trends. The PACF plot exhibits a significant spike at lag 1, followed by smaller spikes, suggesting that an AR component (likely AR(2) or AR(3)) is needed. Additionally, the repeating pattern in the ACF suggests seasonality, meaning a SARIMA model may be more appropriate for capturing both trend and seasonal effects.

Augmented Dickey-Fuller Test:

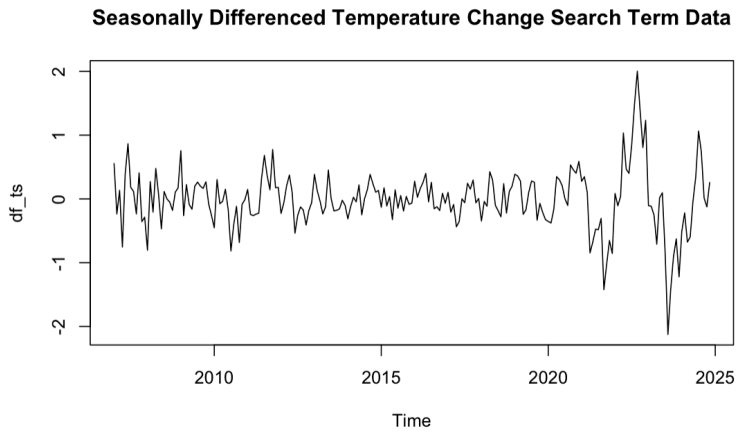
```
Warning: p-value smaller than printed p-value
        Augmented Dickey-Fuller Test

data:  df_ts
Dickey-Fuller = -6.3958, Lag order = 6, p-value = 0.01
alternative hypothesis: stationary
```

The Augmented Dickey-Fuller (ADF) test results indicate stationarity, even though the time series plot may appear non-stationary due to the presence of trends or increasing variance. The apparent trend and increasing variability in the time series plot could be due to heteroscedasticity or seasonality, which the ADF test does not detect. The ADF test confirms that, despite visual trends, the mean-reverting behavior of the series is stationary in a statistical sense. If the variance is changing, transformations like log transformations may be needed for better modeling. Thus, while the series may look non-stationary, it's actually stationary in terms of its underlying statistical properties.

Data Analysis

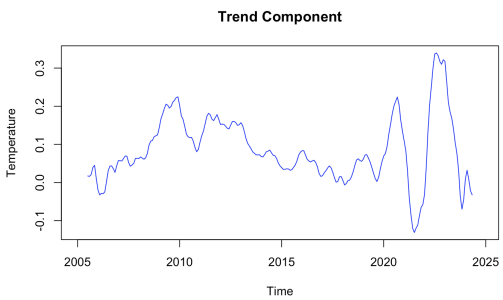


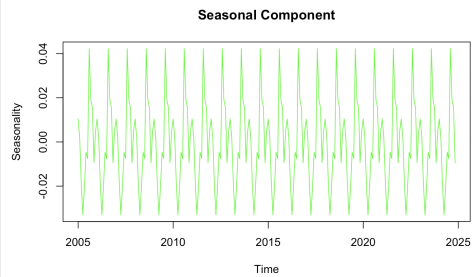
Log Transformation: To stabilize variance and improve interpretability, we applied a log transformation to the time series, addressing heteroscedasticity and making fluctuations more proportional over time. This transformation compressed large variations while expanding smaller values, resulting in a smoother series that better highlights long-term patterns in public interest. While seasonal patterns remain, they appear more proportional, making trend analysis more reliable. This adjustment enhances the ability to detect cyclical behaviors, providing clearer insights into shifts in search interest. If further stationarity is needed, additional transformations like differencing may be considered.

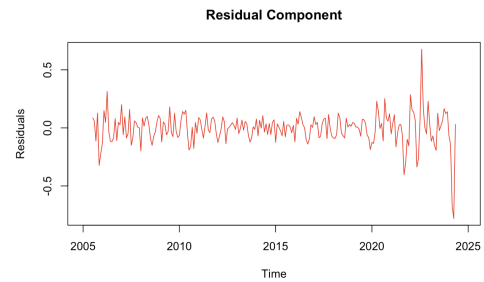


Seasonal Differencing: The seasonally differenced temperature change search term data shows that the long-term trend has been effectively removed, making the

series more stationary. From 2005 to around 2018, fluctuations appear relatively stable, indicating that the seasonal differencing successfully mitigated repetitive seasonal effects. However, after 2019, the variability increases sharply, suggesting potential heteroscedasticity or external influences driving search interest volatility. This could be linked to heightened public discourse on climate change, extreme weather events, or policy changes influencing search behavior. We already applied a log transformation and are unsure of the presence of the sudden spike despite applying differencing, attributing it to other external factors. Further analysis could incorporate additional explanatory variables or techniques to investigate structural breaks to better understand this anomaly.

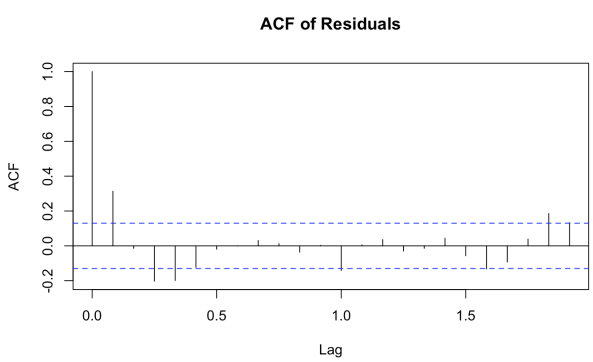


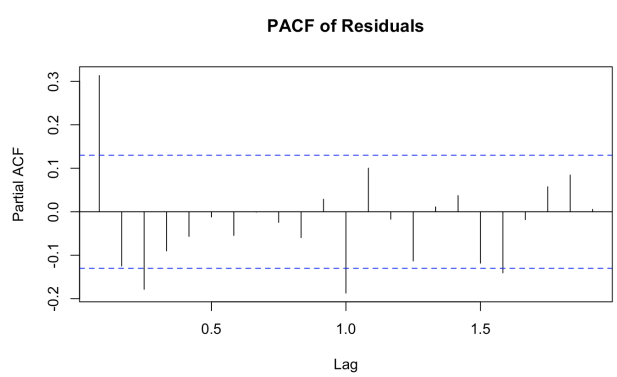




Decomposing the time series helps isolate systematic patterns, allowing better forecasting by modeling trend and seasonality separately. The trend component shows a relatively stable pattern until 2019, after which there is a sharp increase followed by a steep decline, suggesting a potential structural break or external influence affecting the data. The seasonal component remains strong and consistent throughout the time period, indicating a clear recurring pattern, which supports the use of SARIMA modeling to account for seasonality. However, the residual component exhibits increasing volatility after 2019, suggesting heteroscedasticity or missing external factors influencing variability. This increased unpredictability in recent years may require further investigation into external drivers. With these factors in mind, we will first try the ARIMA model and then implement the SARIMA to compare performance.

ACF and PACF of the Residuals for Decomposition:





The ACF plot of the residuals shows that most autocorrelations fall within the significance bounds, indicating that the decomposition effectively removed most structured patterns. However, the slight spike at lag 1 suggests minor short-term autocorrelation, implying that some dependencies remain in the residuals. This could indicate that further refinements, such as incorporating an AR term or testing for stationarity in the residuals, may be necessary.

The PACF plot supports this finding, as most partial autocorrelations are insignificant beyond lag 1, meaning no strong dependencies persist at higher lags. While this suggests the residuals are close to white noise, the minor autocorrelation at lag 1 could indicate an underlying structure that wasn't fully captured. Before fitting predictive models, analyzing ACF and PACF plots helps determine appropriate ARIMA parameters by identifying significant autocorrelation patterns.

To account for these potential issues, we will fit an Arima model and analyze its effectiveness with the box Ljung.

ARIMA model:

```
Call:
arima(x = na.omit(decomp$random), order = c(3, 0, 1))

Coefficients:
         ar1      ar2      ar3      ma1  intercept
      1.1711  -0.4076  -0.0889  -1.0000     -2e-04
s.e.  0.0662   0.1040   0.0742   0.0115      4e-04

sigma^2 estimated as 0.01301:  log likelihood = 168.32,  aic = -324.63

Training set error measures:
                       ME      RMSE        MAE      MPE     MAPE      MASE        ACF1
Training set -0.001991183 0.1140488 0.08083233 95.45544 157.6962 0.7212247 0.003748354
```

The ARIMA(3,0,1) model demonstrates a strong fit with a low AIC (-324.63) and a high log likelihood (168.32), indicating an efficient model selection. The small residual variance (sigma^2 = 0.01301) and near-zero autocorrelation at lag 1 (ACF1 = 0.0037) suggest that the residuals exhibit no significant structure, confirming that the model effectively captures dependencies. Additionally, our low RMSE (0.114) and MAE (0.0808) indicate good predictive accuracy in absolute terms. However, the high MAPE (157.69%) suggests large percentage errors, potentially due to variance fluctuations or extreme values.

Box_Ljung for ARIMA:

```
        Box-Ljung test

data:  residuals(arma_model)
X-squared = 18.332, df = 12, p-value = 0.106
```

The Box-Ljung test results indicate that the p-value is 0.106, which is greater than the typical significance threshold of 0.05. This means we fail to reject the null hypothesis, which states that the residuals exhibit no significant autocorrelation. In other words, the residuals appear to be random (white noise), suggesting that the ARMA model has adequately captured the underlying structure of the time series. Although a slight autocorrelation is present at lag 1, the Box-Ljung test confirms that there is no significant overall autocorrelation in the residuals. This suggests that our ARMA model sufficiently captures the underlying structure of the time series, and no major adjustments are necessary. We will now try the Sarima model for potential improvements.

SARIMA Model:

```
ARIMA(3,0,1)(1,1,1)[12]

Coefficients:
          ar1     ar2      ar3     ma1     sar1     sma1
      -0.0286  0.2469  -0.2511  0.6676  -0.2705  -0.9606
s.e.   0.2343  0.1477   0.0746  0.2334   0.0814   0.1595

sigma^2 = 0.02386:  log likelihood = 87.53
AIC=-161.06   AICc=-160.55   BIC=-137.08

Training set error measures:
                      ME      RMSE       MAE MPE MAPE      MASE       ACF1
Training set 0.006502496 0.1485363 0.1027024 NaN  Inf 0.5494844 0.02316967
```

The SARIMA(3,0,1)(1,1,1)[12] model includes three autoregressive (AR) terms, one moving average (MA) term, and seasonal components. The log likelihood is 87.53, with an AIC of -161.06, indicating a weaker fit compared to ARIMA. The RMSE (0.148) and MAE (0.1027) suggest slightly higher prediction errors, while the MAPE is infinite (Inf), likely due to extreme percentage errors. The ACF1 value (0.0232) indicates low residual autocorrelation, meaning the model captures most of the dependencies. However, despite producing white noise residuals, SARIMA does not significantly improve over ARIMA in terms of overall model accuracy.

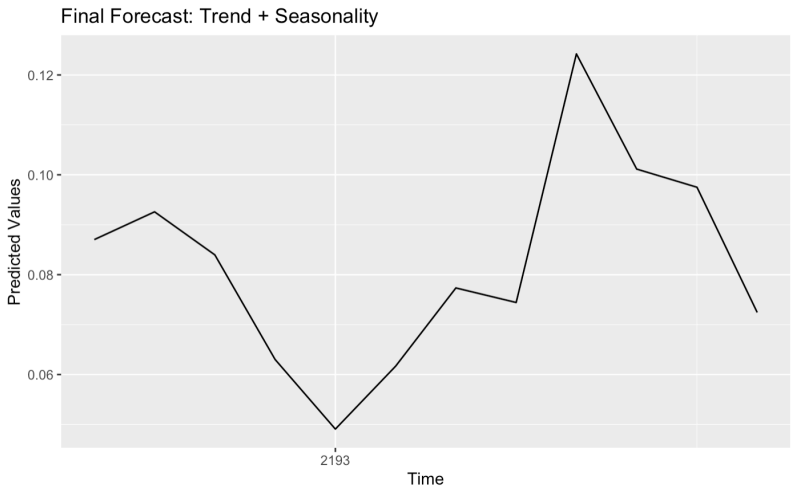
Box-Ljung test for SARIMA

```
        Box-Ljung test

data:  residuals(sarima_model)
X-squared = 12.563, df = 12, p-value = 0.4016
```

The Box-Ljung test for the SARIMA model residuals yields a p-value of 0.4016, which is well above the typical significance threshold of 0.05. This means we fail to reject the null hypothesis, indicating that there is no significant autocorrelation in the residuals. In other words, the SARIMA model has effectively captured the dependencies in the time series, leaving behind white noise residuals.

The SARIMA(3,0,1)(1,1,1)[12] model incorporates both seasonal and non-seasonal components, but its performance metrics suggest that it is less effective than ARIMA(3,0,1). The log likelihood (87.53) and AIC (-161.06) are significantly worse than those of ARIMA, indicating a weaker model fit. While the residual autocorrelation (ACF1 = 0.0232) is still low, suggesting independent residuals, the RMSE (0.148) and MAE (0.1027) are higher than in ARIMA, meaning its predictions are less accurate. Additionally, the MAPE is infinite (Inf), likely due to extreme errors in percentage-based calculations, which suggests instability in the model's predictive ability. Given these results, SARIMA does not provide a meaningful improvement over ARIMA + Seasonal Forecasting, and it may not be necessary to include in the final analysis.

Forecast: Trend and Seasonality



The final forecast, which combines predicted trend and seasonal components, successfully captures the recurring seasonal patterns observed in the data. The forecasted values exhibit fluctuations, with an initial decline followed by a sharp increase and a subsequent drop,

reinforcing the strong seasonal influence on the time series. This suggests that historical seasonal cycles remain a dominant factor in predicting future values. While the model effectively integrates both trend and seasonality, the sharp variations toward the end may indicate potential volatility or external influences not accounted for. Overall, the forecast provides useful short-term predictions, but further validation with accuracy metrics like RMSE and consideration of external factors could enhance the reliability of the model.

```
[1] "Residual RMSE: 0.114048849777332"
```

```
[1] "Forecast RMSE: 0.506151035917661"
```

The forecast RMSE of 0.506 suggests that the ARIMA(3,0,1) model has moderate prediction accuracy when tested on the last 12 months of data. Compared to the residual RMSE of 0.114, the forecast RMSE is significantly higher, indicating that the model performs well on training data but struggles more with generalization to unseen future values. This could suggest potential overfitting to past trends or a need for improved seasonal modeling.

Discussion

The time series analysis of Google Trends data for “temperature change” revealed key insights into public search behavior, along with several challenges in the modeling process. Our initial decomposition identified a clear upward trend and strong seasonality, reinforcing the need for models that account for both long-term patterns and periodic fluctuations. The Augmented Dickey-Fuller test confirmed stationarity, but visual inspection revealed persistent large fluctuations, particularly after 2019, suggesting increased volatility that standard differencing did not fully address. This raises concerns about heteroscedasticity or external influences, such as heightened public discourse, extreme weather events, or media-driven spikes, which our models did not explicitly capture.

The ARIMA(3,0,1) model demonstrated a strong in-sample fit with a low AIC (-324.63), low residual RMSE (0.114), and near-zero residual autocorrelation (ACF1 = 0.0037), confirming that it effectively captured the time series structure. However, despite this strong performance on historical data, the out-of-sample forecast RMSE increased to 0.506, indicating that while ARIMA models past data well, it struggles with generalization. The presence of persistent fluctuations in residuals suggests that the model might be missing key external variation factors, leading to increased prediction errors. While the ARIMA model demonstrated strong in-sample fit (low AIC, low residual variance, and white noise residuals), it struggled with generalization in

later years due to increased volatility. However, it successfully captured long-term trends and seasonal cycles, making it a useful tool for short-term forecasting.

The SARIMA(3,0,1)(1,1,1)[12] model incorporated seasonality directly but failed to outperform ARIMA. While its residuals were white noise (Box-Ljung p-value = 0.4016), its AIC (-161.06) was significantly worse than ARIMA, and its RMSE (0.148) was higher, indicating lower predictive accuracy. Additionally, the constant presence of large fluctuations remained, even after applying seasonal adjustments, suggesting that seasonal differencing alone was insufficient to stabilize variance. The MAPE value was infinite (Inf), further highlighting potential instability in percentage-based error calculations. Given these results, SARIMA did not provide a meaningful improvement over ARIMA + Seasonal Forecasting, and including it does not significantly enhance forecasting performance.

Ultimately, the final forecast, which combined predicted trend and seasonal components, effectively captured recurring seasonal patterns but continued to exhibit sharp, unpredictable variations. The fluctuations which have been present throughout the dataset suggest that external influences are at play. These include things such as climate events, public awareness campaigns, or policy changes and they may play a significant role in shaping search interest. The higher forecast RMSE suggests the need for additional refinements, such as incorporating external explanatory variables (e.g., climate indices, news sentiment analysis) or employing models designed for handling variance instability. Despite these limitations, this study demonstrates how online search trends can serve as a valuable indicator of public engagement with climate-related topics, while also highlighting the challenges of forecasting highly dynamic, socially influenced phenomena.

Conclusion

This study applied time series decomposition, ARIMA, and SARIMA modeling to analyze Google Trends data on search interest in “temperature change” and its predictability. The results confirm a strong seasonal pattern, indicating that public interest in temperature change is cyclical and influenced by recurring climate events. ARIMA modeling was effective in capturing historical trends but struggled with forecast accuracy, as indicated by the higher forecast RMSE (0.506) compared to residual RMSE (0.114). While SARIMA was tested to model seasonality explicitly, it did not outperform ARIMA + Seasonal Forecasting, making it an unnecessary addition to the final model.

The findings highlight both the potential and limitations of using online search data for climate-related forecasting. While historical patterns are well captured, the presence of unmodeled external influences—such as climate events, public policy changes, and media-driven discourse—introduces forecasting uncertainty. Future studies could improve accuracy by

incorporating external variables (e.g., temperature anomalies, climate reports, or policy shifts) and testing alternative models that address heteroscedasticity. Despite its limitations, this study demonstrates that search interest data provides valuable insights into public engagement with climate change, underscoring the growing intersection between digital behavior and environmental awareness.

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STA 137 Final Project

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2025-03-11

R Appendix

```r
knitr::opts_chunk$set(
  echo=FALSE,
  results="hide",
  warning=FALSE,
  message=FALSE,
  fig.show="hide"
)
library(zoo)
library(tseries)
library(forecast)
library(ggplot2)
library(dplyr)
df <- read.csv("/Users/judynei/Downloads/multiTimeline.csv",
               stringsAsFactors = FALSE,
               skip = 2)  # Skip first 2 rows
colnames(df) <- c("Month", "Temp")
# Find outliers using boxplot.stats
outlier_values <- boxplot.stats(df$Temp)$out

# Count of outliers
length(outlier_values)

# Identify the lower and upper bounds for outlier detection
Q1 <- quantile(df$Temp, 0.25)
Q3 <- quantile(df$Temp, 0.75)
IQR <- Q3 - Q1
lower_bound <- Q1 - 1.5 * IQR
upper_bound <- Q3 + 1.5 * IQR

# Removing outliers using explicit condition
df_clean <- df[df$Temp >= lower_bound & df$Temp <= upper_bound, ]

# Boxplots before and after removing outliers
par(mfrow = c(1, 2))

boxplot(df$Temp, main = "Boxplot of Search Interest (Before Removing Outliers)",
        ylab = "Temperature", col = "lightblue")
boxplot(df_clean$Temp, main = "Boxplot of Search Interest (After Removing Outliers)",
```

```r
        ylab = "Temperature", col = "lightgreen")
ggplot(df, aes(x = Temp)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Histogram of Search Interest",
       x = "Search Interest",
       y = "Frequency") +
  theme_minimal()
# Convert to time series object (monthly data, starts in January 2004)
df_ts <- ts(na.omit(df_clean$Temp), start = c(2004, 1), frequency = 12)

summary(df_ts)

# Perform Augmented Dickey-Fuller Test for stationarity
adf_result <- adf.test(df_ts)

# Print ADF test results
print(adf_result)

# Plot the raw time series
plot(df_ts, type = "l", col = "blue",
     main = "Raw Monthly Temperature Change Search Interest Time Series",
     xlab = "Time", ylab = "Search Interest")
library(ggplot2)
library(zoo)


rolling_window <- 12  # Adjust window size as needed

# Compute rolling mean and rolling standard deviation
rolling_mean <- rollmean(df_ts, k = rolling_window, fill = NA, align = "right")
rolling_sd <- rollapply(df_ts, width = rolling_window, FUN = sd, fill = NA, align = "right")

# Create a data frame for plotting
rolling_stats_df <- data.frame(
  Time = time(df_ts),
  Rolling_Mean = rolling_mean,
  Rolling_SD = rolling_sd
)

# Plot rolling mean and rolling standard deviation together
ggplot(rolling_stats_df, aes(x = Time)) +
  geom_line(aes(y = Rolling_Mean, color = "Rolling Mean"), size = 1) +
  geom_line(aes(y = Rolling_SD, color = "Rolling Std Dev"), size = 1) +
  scale_y_continuous(
    name = "Rolling Mean",
    sec.axis = sec_axis(~., name = "Rolling Std Dev")  # Secondary y-axis
  ) +
  labs(title = "Rolling Mean and Standard Deviation",
       x = "Time") +
  scale_color_manual(values = c("Rolling Mean" = "blue", "Rolling Std Dev" = "red")) +
  theme_minimal() +
  theme(legend.title = element_blank())
par(mfrow = c(1,2))
```

```r
acf(df_ts, main = "ACF of Search Interest")
pacf(df_ts, main = "PACF of Search Interest")
df_ts <- log(df_ts)
plot(df_ts, type = "l", col = "blue", main = "Log Transformed Time Series")
df_ts <- diff(df_ts, lag = 12)  # monthly differencing
plot(df_ts, main = "Seasonally Differenced Temperature Change Search Term Data")
# Decompose the time series
decomp <- decompose(df_ts, type = "additive")

plot(decomp$trend, main = "Trend Component", col = "blue", ylab = "Temperature")
plot(decomp$seasonal, main = "Seasonal Component", col = "green", ylab = "Seasonality")
plot(decomp$random, main = "Residual Component", col = "red", ylab = "Residuals")
acf(na.omit(decomp$random), main = "ACF of Residuals")
pacf(na.omit(decomp$random), main = "PACF of Residuals")
arma_model <- arima(na.omit(decomp$random), order = c(3, 0, 1))
summary(arma_model)

Box.test(residuals(arma_model), lag = 12, type = "Ljung-Box")

# Decompose time series
decomp <- decompose(df_ts)

# Extract components
trend <- decomp$trend
seasonality <- decomp$seasonal
residuals <- na.omit(decomp$random)  # Remove NAs from residuals

# Forecast trend using linear model
trend_ts <- ts(na.omit(trend), frequency = 12)  # Adjust frequency if needed
trend_model <- tslm(trend_ts ~ trend)  # Linear trend model
trend_forecast <- forecast(trend_model, h = 12)  # Forecast 12 months ahead

# Extend seasonality by repeating past cycles
seasonal_pattern <- tail(seasonality, 12)  # Last year's seasonality
seasonal_forecast <- rep(seasonal_pattern, length.out = 12)  # Repeat

# Plot trend forecast
autoplot(trend_forecast) + ggtitle("Trend Forecast")

# Combine trend and seasonal forecasts
final_forecast <- trend_forecast$mean + seasonal_forecast

# Convert to time series for visualization
final_forecast_ts <- ts(final_forecast, start = end(df_ts)[1] + c(0,1), frequency = 12)

# Plot the final forecast
autoplot(final_forecast_ts) +
  ggtitle("Final Forecast: Trend + Seasonality") +
  xlab("Time") + ylab("Predicted Values")
# Compute RMSE using residuals of ARIMA model
residual_rmse <- sqrt(mean(residuals(arma_model)^2, na.rm = TRUE))
print(paste("Residual RMSE:", residual_rmse))
train_ts <- window(df_ts, end = c(2023, 12))
```

```r
test_ts <- window(df_ts, start = c(2024, 1))

arma_model <- arima(train_ts, order = c(3,0,1))
forecast_values <- forecast(arma_model, h = 12)

predicted_values <- forecast_values$mean

library(Metrics)

# Compute RMSE
rmse_value <- rmse(test_ts, predicted_values)
print(paste("Forecast RMSE:", rmse_value))
sarima_model <- Arima(df_ts, order = c(3,0,1), seasonal = c(1,1,1)) # (3,1) order
summary(sarima_model)

Box.test(residuals(sarima_model), lag = 12, type = "Ljung-Box")
```